

APTIVA

Intelligence that adapts to you.

Task 5 — UI Design and Implementation

UI Design Document

App Identity · Visual Design · Frontend Implementation

Project	Design and Implementation of an Adaptive E-Learning Platform based on Quality of Experience for Inclusive Online Education in Low-Resource Settings
App Name	Aptiva
Tagline	"Intelligence that adapts to you."
Course	CEF 440: Internet Programming and Mobile Programming
Institution	University of Buea — Faculty of Engineering and Technology
Department	Department of Computer Engineering
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Document Version	v1.0
Date	May 2025

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Part 1: App Identity

1.1 App Name and Brand Rationale

Application Name: Aptiva

Tagline: *"Intelligence that adapts to you."*

The name Aptiva was engineered from the Latin root "**aptus**" meaning 'fitted, adapted, suitable' — combined with the suffix "**-iva**" connoting active, ongoing intelligent behaviour. The result communicates three things instantly:

- Adaptation — the platform dynamically fits itself to the student's current network reality
- Intelligence — Aptiva uses ML recommendation, QoE feedback loops, and adaptive algorithms — not just static content delivery
- Brevity — three syllables, works in English and French, no translation required for Cameroonian students

BRAND	Aptiva is not just an e-learning app. It is an intelligent adaptive learning intelligence that removes the barrier of connectivity from education. The brand communicates confidence, innovation, and technological leadership.
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1.2 Visual Identity System

1.2.1 App Icon

The Aptiva icon is built for recognition at 48×48dp on low-resolution screens and premium recognition at 1024×1024px on app store listings.

Design elements:

- Background: Deep space navy (#0A0E1A) — communicates sophistication, technology, depth

- Primary mark: Electric cyan diamond (◆) — the core Aptiva geometric symbol — representing precision, adaptability, and faceted intelligence
- Lettermark: 'APTIVA' in tight letter-spacing beneath the diamond in electric cyan
- Glow effect: Soft cyan radial glow around the diamond — communicates energy and intelligence
- The diamond form was chosen because it has no single orientation — it adapts to any rotation — directly mirroring Aptiva's adaptive philosophy

1.2.2 Logo Variants

Variant	Usage	Colours	Dimensions
Primary Icon	App icon, splash screen	Navy background, Cyan diamond	1024x1024px (store), 512x512px (web)
Full Wordmark	Documents, presentations	Transparent background, Cyan + White	SVG scalable
Light Wordmark	Dark backgrounds	White container, Cyan mark	Any
Dark Wordmark	Light backgrounds	Navy container, White mark	Any
Monochrome	Print, grayscale	All black or all white	Any

1.3 Aptiva Color System

The Aptiva color palette is a futuristic dark-first system. Unlike traditional light-mode apps, Aptiva uses a deep navy base with electric accent colours. This design decision was deliberate:

- Dark backgrounds reduce screen brightness consumption — critical for students with battery-constrained low-end Android devices
- High-contrast neon accents on dark backgrounds read clearly on low-quality LCD screens

- The palette communicates technological sophistication and modernity — appropriate for an AI-powered platform

Swatch	Role	Name	Hex	Usage
#00D4FF	Primary Accent	Electric Cyan	#00D4FF	Primary buttons, links, active states, progress bars, borders, section indicators
#6366F1	Secondary Accent	Electric Indigo	#6366F1	Instructor features, AI badge, secondary highlights, charts
#39D98A	Success	Electric Lime	#39D98A	Excellent signal badge, correct quiz answers, upload complete, sync success
#FBBF24	Warning	Amber	#FBBF24	Good signal badge, star ratings, caution states, 'Good' quality indicator
#F43F5E	Danger / Alert	Electric Rose	#F43F5E	Audio-only mode banner, <30% completion alert, error states, 'Poor' signal
#A855F7	AI / Recommendation	Violet	#A855F7	AI recommendation badge, recommendation engine outputs, ML-related features
#0A0E1A	Background	Space Navy	#0A0E1A	Primary screen background — dark base of entire visual system
#111827	Surface 1	Dark Slate	#111827	App bar, bottom navigation, card backgrounds at lowest elevation

#1A2236	Surface 2	Card Dark	#1A2236	Standard card background — elevated above primary background
#1E2D45	Border	Dark Border	#1E2D45	Subtle borders, dividers, input field outlines on dark background
#F8FAFC	Text Primary	Near White	#F8FAFC	All primary body text, headings, main readable content
#94A3B8	Text Secondary	Silver	#94A3B8	Secondary information, subtitles, metadata, captions
#64748B	Text Muted	Slate	#64748B	Placeholder text, disabled states, tertiary information

1.4 Typography System

Aptiva uses two typefaces that provide maximum legibility on low-resolution screens while communicating technological sophistication:

Style	Font	Weight	Size	Flutter TextStyle	Usage
Display	Poppins	Bold 700	32sp	displayLarge	Aptiva wordmark on splash
Heading 1	Poppins	Bold 700	24sp	headlineLarge	Screen titles
Heading 2	Poppins	SemiBold 600	20sp	headlineMedium	Section headers, card titles
Heading 3	Poppins	SemiBold 600	18sp	headlineSmall	Subsection labels, dialog titles

Body Large	Inter	Regular 400	16sp	bodyLarge	Primary content, course descriptions
Body Medium	Inter	Regular 400	14sp	bodyMedium	Secondary content, list items
Body Small	Inter	Regular 400	12sp	bodySmall	Captions, metadata, timestamps
Label	Inter	Medium 500	12sp	labelMedium	Buttons, chips, badges
Mono	Courier New	Regular 400	12sp	Custom	Network metrics, technical values

1.5 Iconography

- Icon library: Material Design Icons (material_design_icons_flutter package)
- Icon style: Rounded corners — consistent with Material Design 3 / Material You
- Standard size: 24dp action icons; 20dp inline icons; 48dp feature icons; 72dp illustrative icons
- Network quality uses custom Lottie JSON animations for smooth state transitions
- All icons must have semanticsLabel for VoiceOver / TalkBack screen reader accessibility
- Icon colour always matches the semantic colour of its context (cyan for primary, rose for danger, etc.)

1.6 Design Principles

Principle	Description	How Implemented in Aptiva
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Dark-First Intelligence	Dark backgrounds as the default — not an option	Every screen uses Space Navy (#0A0E1A) base — optimised for OLED screens and low brightness
Signal Transparency	Network quality always visible	Persistent network quality badge on dashboard header and video player — never hidden
Progressive Disclosure	Show minimum; reveal on demand	Collapsed descriptions, bottom sheets for detail, accordion quiz sections
Adaptive Communication	UI communicates current mode clearly	Rose banner for audio-only, green badge for excellent, offline icons for cached content
Accessible by Default	Readability as a non-negotiable baseline	48dp touch targets, 4.5:1 contrast, 12sp minimum, WCAG 2.1 AA everywhere
Futuristic Restraint	Innovation without overwhelming	Max 3 accent colours per screen; no gradients in content areas; clean geometric forms

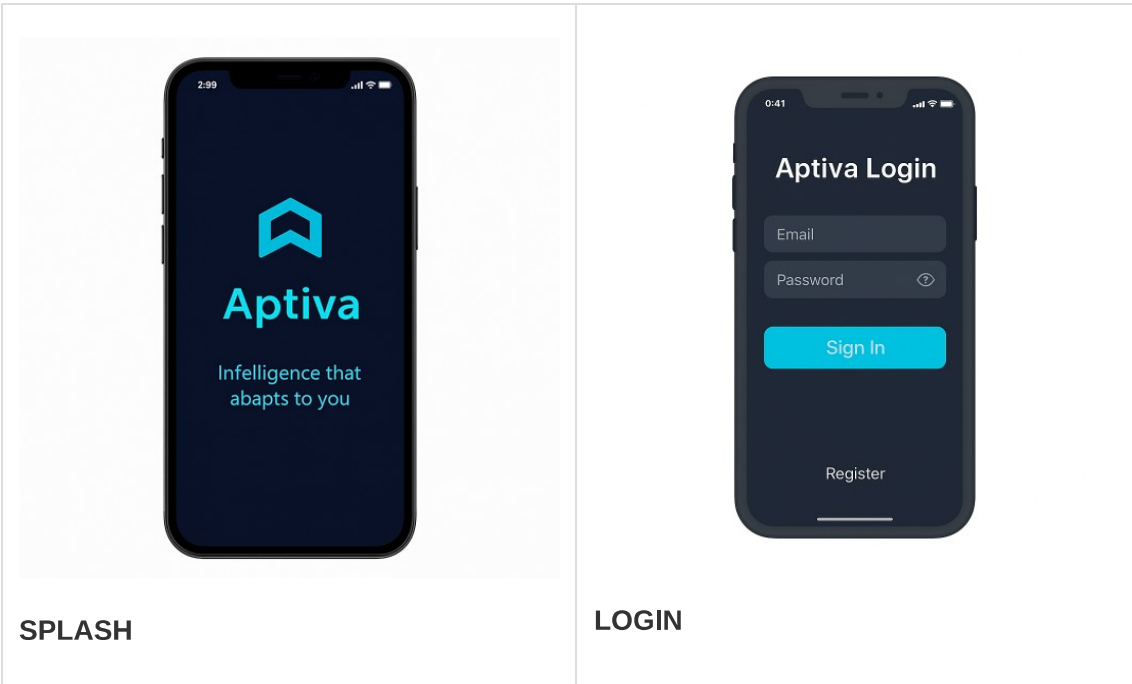
Part 2: Visual Design — Wireframes

The following wireframes define the complete visual specification for Aptiva. All screens are designed to **390×844px** — the standard Flutter device canvas equivalent to iPhone 14 / Samsung Galaxy S21. Every measurement, colour reference, spacing value, and component behaviour described below is binding for the Flutter implementation in Part 3.

NOTE	Aptiva uses a dark-first design system. All wireframes are rendered on the Space Navy (#0A0E1A) dark background. This is not a dark mode option — it is the primary and only visual theme of the application.
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2.1 Authentication Flow — Screens 1, 2, 3

Screens 1 and 2: Splash Screen and Login



Splash Screen Specification

Element	Value / Behaviour	SRS Ref
Background	Full-screen Space Navy (#0A0E1A) — url(#bgGrad) linear from #0D1526 to #0A0E1A	App Identity 1.3
Glow circles	Three concentric cyan circles (r=140, 100, 60dp) with decreasing opacity — creates depth illusion	App Identity 1.2
App icon	Cyan diamond (◆) 42sp centred in innermost circle	App Identity 1.2
App name lettermark	'APTIVA' in 10sp all-caps, Cyan, letter-spacing 4px — identity mark below diamond	App Identity 1.1
App name wordmark	'Aptiva' Poppins Bold 34sp White — main display title	App Identity 1.4

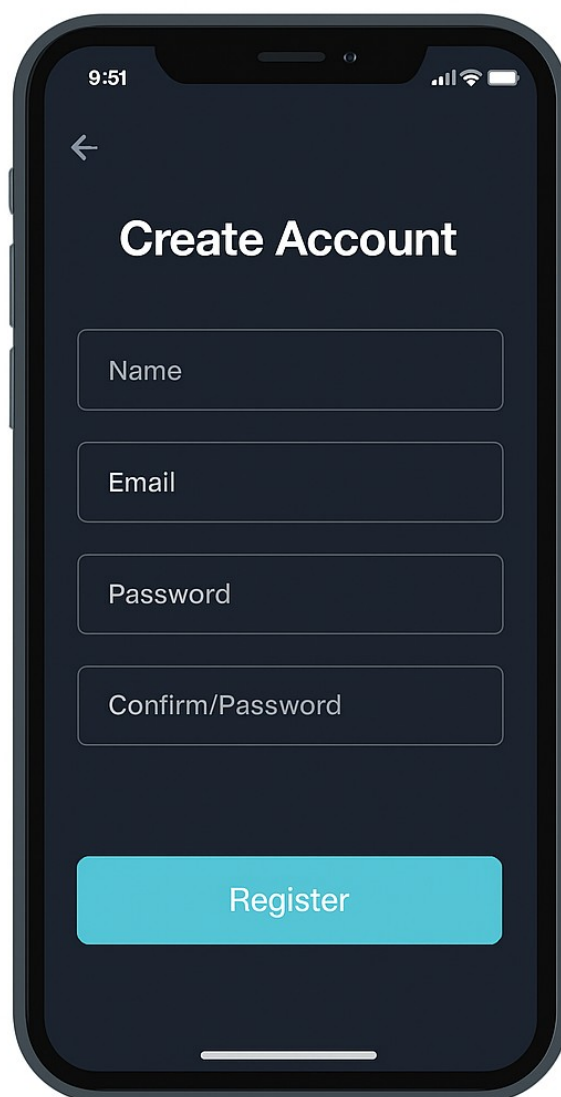
Tagline	'Intelligence that adapts to you.' Inter Regular 13sp Silver	App Identity 1.1
Progress bar	120dp wide; dark navy background; cyan fill animates left-to-right over 2.5 seconds	NFR-PERF-07
Version label	'v1.0.0' 9sp Slate — bottom of screen	Technical
Duration	Displays 2.5 seconds then routes: authenticated user → Dashboard; unauthenticated → Login	FR-AUTH-02

Login Screen Specification

Element	Value / Behaviour	SRS Ref
Background	Space Navy with subtle cyan glow circle top-left; indigo glow top-right	App Identity 1.3
App name	'APTIVA' lettermark 13sp Cyan letter-spacing 4px — centred at y=100	App Identity 1.1
Title	'Welcome back' Poppins Bold 28sp White centred	App Identity 1.4
Subtitle	'Sign in to continue' Inter Regular 13sp Silver centred	App Identity 1.4
Email field	Label 10sp Slate; Card background #1A2236; 48dp height; 10dp radius; focused border = Cyan 1dp	FR-AUTH-02
Password field	Same style as email; show/hide toggle (eye icon) right side	FR-AUTH-02
Forgot password	Right-aligned 11sp Cyan hyperlink	FR-AUTH-02

Sign In button	Full-width 50dp; Cyan fill; Navy text Poppins Bold 14sp; 10dp radius	FR-AUTH-02
Google Sign-In	Card background; dark border 0.5dp; Google 'G' dark circle left; 'Sign in with Google' Silver	FR-AUTH-01
Register link	'Don't have an account?' Muted + 'Create account' Cyan Bold below	FR-AUTH-01

Screen 3: Register

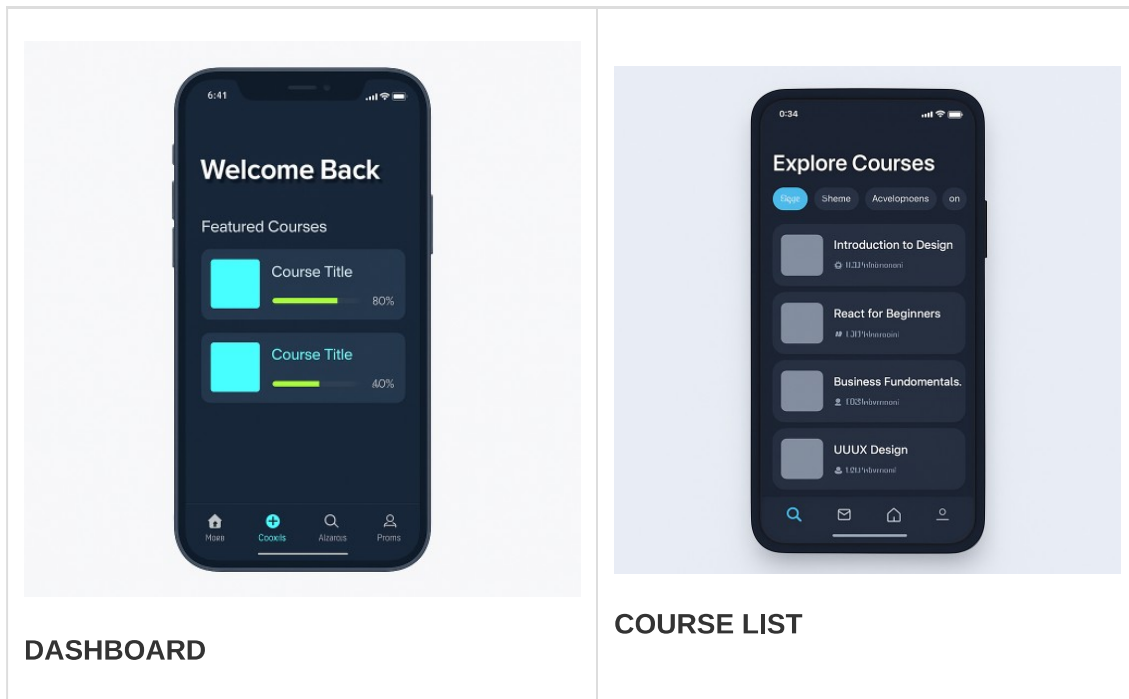


REGISTER SCREEN

Element	Value / Behaviour	SRS Ref
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Title	'Create Account' Poppins Bold 26sp White — top-left; Cyan glow circle top-right decoration	FR-AUTH-01
Name / Email / Password	Same dark input style as Login; password shows strength indicator below	FR-AUTH-01
Role Selector	Two-button toggle: 'Student' active=Cyan fill Navy text; 'Instructor' inactive=Card border	FR-AUTH-03
Create Account button	Full-width Cyan button — same style as Sign In	FR-AUTH-01
Terms link	'Terms of Service & Privacy Policy' Indigo underlined hyperlink	Legal
Input validation	Real-time rose-coloured error text below each invalid field; green checkmark when valid	FR-AUTH-01

2.2 Student Core Learning Flow — Screens 4 and 5



Student Dashboard Specification

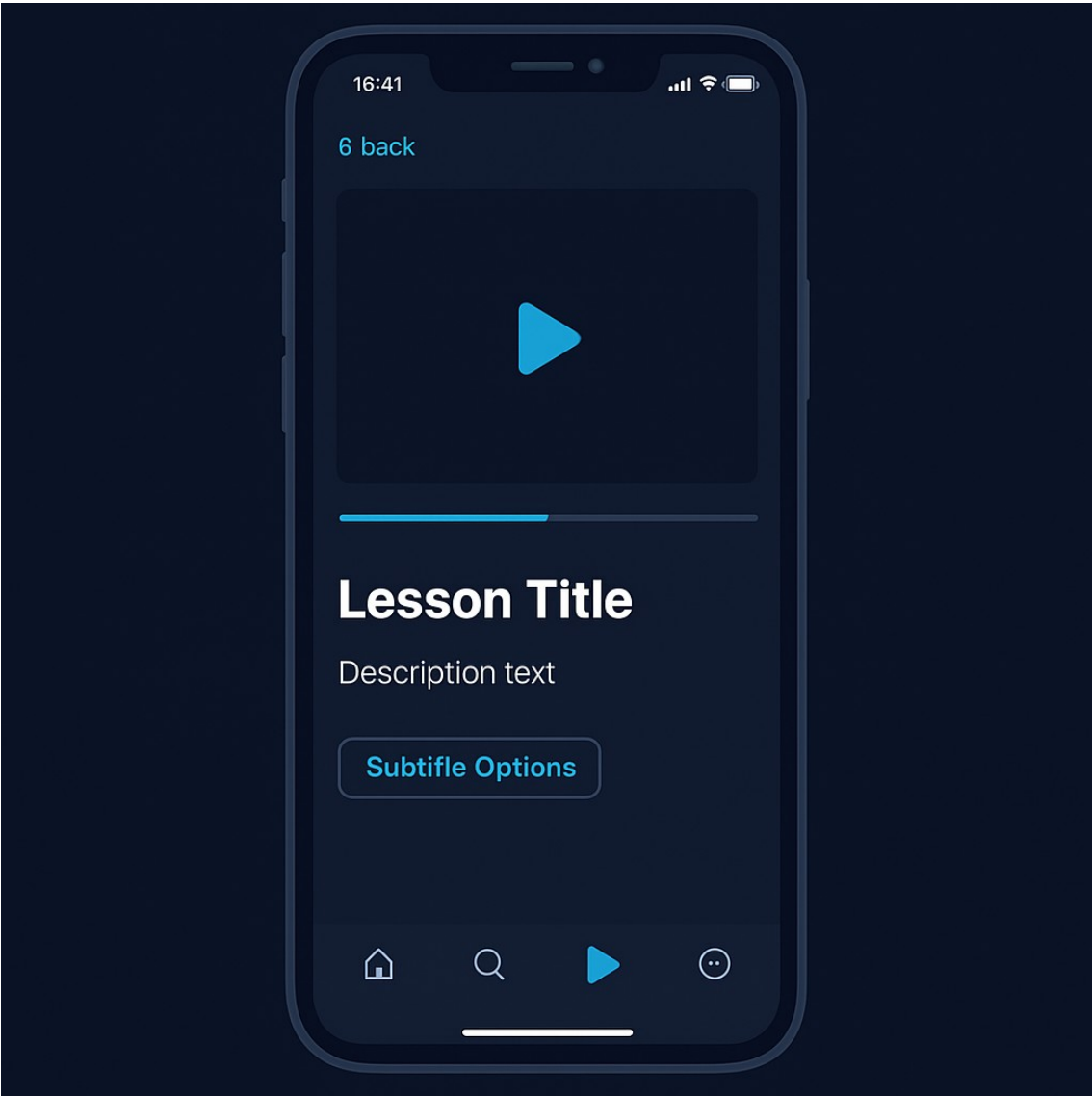
Element	Value / Behaviour	SRS Ref
Status bar	32dp Space Navy background; Silver system text	System
App bar	Surface 1 (#111827) 96dp; greeting Poppins Bold 15sp White; subtitle 11sp Silver; cyan bottom border 0.5dp	FR-AUTH-03
Network badge	Persistent pill badge: Lime green=Excellent Amber=Good Rose=Poor Slate=Offline — ALWAYS visible	FR-NET-02
Continue Learning card	Dark card with cyan top accent bar 4dp; course name + Cyan progress bar + Play button	FR-COURSE-03
My Courses grid	2-column; course cards with dark thumbnail area; title Poppins Bold 12sp; instructor 10sp Silver; Cyan progress bar	FR-COURSE-03
Recommended by AI	Full-width card; Violet left area with star icon; violet 'Recommended' pill badge; Violet AI badge top-right	FR-REC-01
Today stats row	3 cards with large coloured numbers: Cyan=watch time, Lime=quizzes, Violet=score	FR-ANALYTICS-01
Bottom navigation	Surface 1 72dp; Cyan active tab with top indicator bar 3dp; icons + labels	App Identity 1.6

Course List Specification

Element	Value / Behaviour	SRS Ref
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Search bar	Rounded 44dp Card background; dark border 0.5dp; hint text 13sp Muted	FR-COURSE-03
Filter chips	Horizontal scroll: All (active=Cyan fill)/Video/Audio/Offline — inactive=Card + dark border	FR-COURSE-04
Course card	136dp; dark thumbnail left 90dp with format circle icon; title Bold 13sp; instructor 10sp Silver; format badge pill	FR-COURSE-04
Format badges	HD=Cyan Audio=Rose Offline=Slate — pill with matching background at 22% opacity	FR-COURSE-04
Progress bar	5dp height; border background; Cyan fill; percentage text 10sp Muted	FR-COURSE-03

2.3 Video Player Screen — Screen 6



VIDEO PLAYER SCREEN

Element	Value / Behaviour	SRS Ref
Video viewport	Full-width; dark Surface background (#111827); cyan glow circle overlay on play button	FR-ADAPT-01
Quality badge	Persistent pill top-left: 'HD 480p' Lime green 'SD' Amber 'Audio Only' Rose — changes with band	FR-NET-02
Progress scrubber	3dp cyan bar; white circle handle 12dp; taps to seek; time stamps both ends 11sp Muted	FR-COURSE-03

Playback controls	5 controls: << << >> >> ; pause icon in Cyan larger 24sp; others Silver 18sp	FR-COURSE-03
Lecture info card	Dark card; title Poppins SemiBold 12sp; course + instructor Inter 10sp Muted	FR-COURSE-03
Lecture list	Scrollable; active lecture Cyan border 1.5dp + Cyan background tint; 'Playing' Cyan badge; 68dp per item	FR-COURSE-03
Network monitor panel	Dark card at bottom; band badge + live bandwidth value + quality switch threshold info	FR-NET-01

2.4 Audio-Only Mode — Screen 7 (Critical Adaptive Screen)



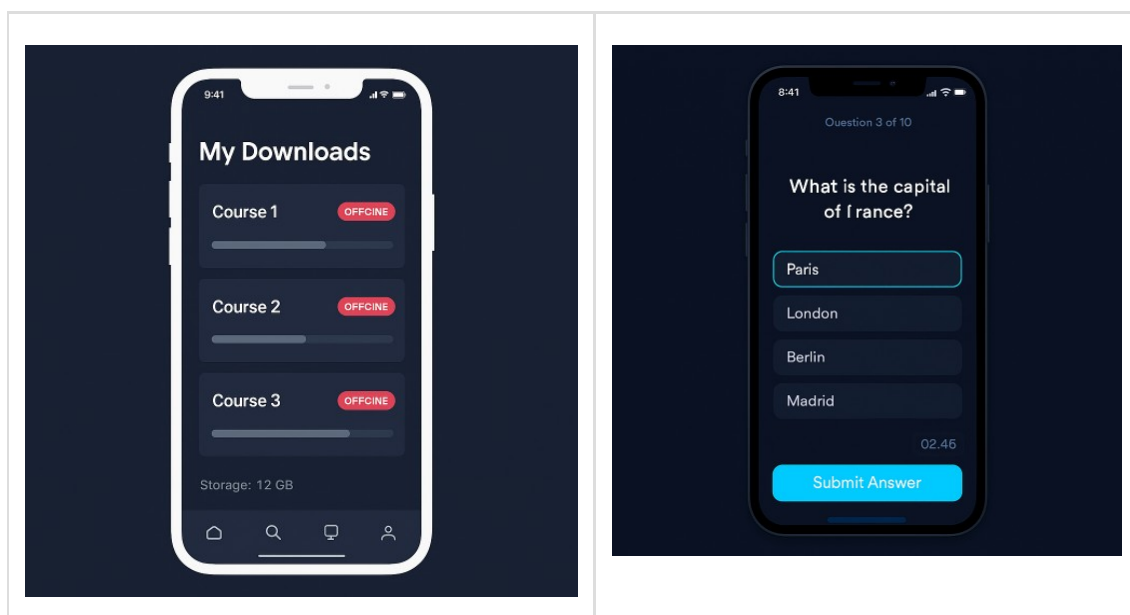
AUDIO-ONLY MODE

Design priority: This screen activates automatically in under 2 seconds when the QoS Classifier downgrades to Poor band. The design must **reassure** the student — learning has not stopped. The audio-only mode is a feature, not a failure.

Element	Value / Behaviour	SRS Ref
Header	Dark Surface app bar with back arrow; title 'Now Playing' White Bold; subtitle 'Low bandwidth mode' Muted	FR-ADAPT-02

Alert banner	56dp Rose (#F43F5E) 22% opacity background; Rose border top and bottom; 'Low bandwidth detected' Rose Bold 13sp; subtitle Rose 10sp	FR-ADAPT-02
Audio visualiser area	Full-width Surface (#111827) dark panel 300dp height; three nested cyan glow circles with headphone icon	FR-ADAPT-02
Audio waveform	8 animated Rose vertical bars with rounded ends; heights animate to audio amplitude in real time	FR-ADAPT-02
Scrubber + controls	Rose fill for progress bar (visually communicates degraded state); Amber pause icon for visibility	FR-ADAPT-01
Bandwidth restored card	Dark card; Lime checkmark circle; 'Bandwidth restored above 500 kbps' White Bold 12sp; 'Tap to resume video' Cyan link; Lime 'Resume' button	FR-ADAPT-02
Network status panel	Dark card; 'NETWORK STATUS' 11sp Muted uppercase; live Bandwidth/Latency/Packet Loss/Status values; Status in Rose when audio-only	FR-NET-01

2.5 Offline Content and Quiz — Screens 8 and 9



OFFLINE	QUIZ
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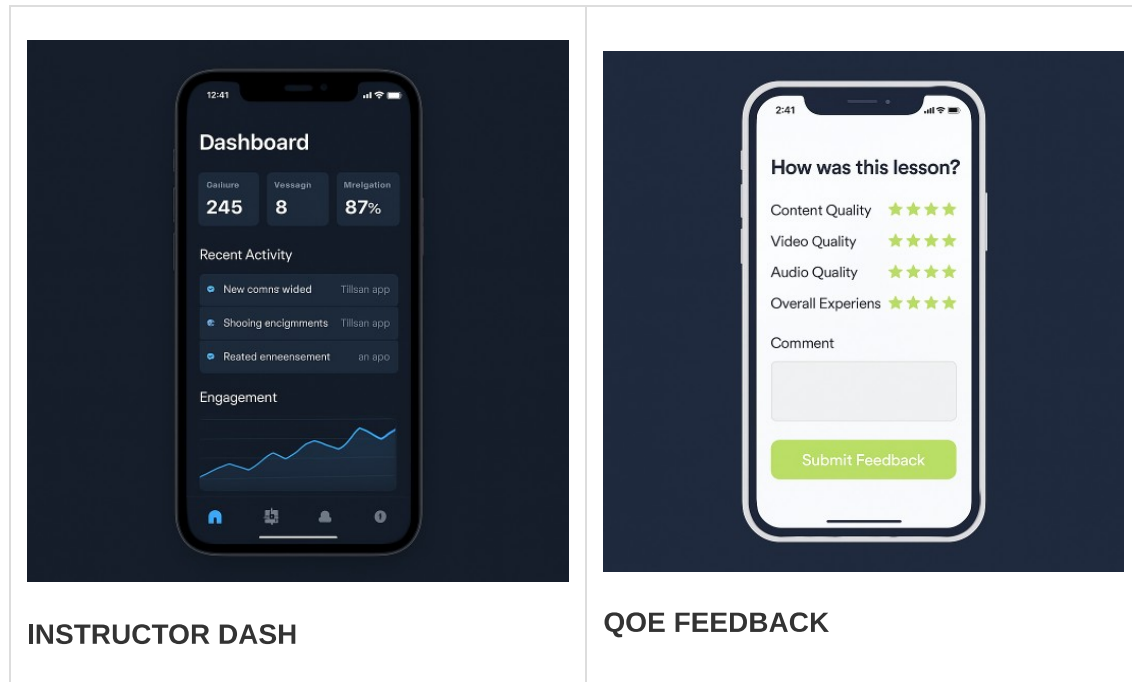
Offline Content Screen Specification

Element	Value / Behaviour	SRS Ref
Storage bar	Cyan fill linear bar over dark background; used/total GB text; 'Manage' Cyan link	FR-ADAPT-03
Sync status	Last synced timestamp; Lime 'Online' pill when connected; Rose 'Offline' pill when not	FR-ADAPT-03
Download cards	116dp height; dark thumbnail left with Cyan 'Offline' badge; play circle; progress bar; watch %	FR-ADAPT-03
Empty state	If no downloads: Cyan cloud-download icon; 'No offline content' heading; 'Browse courses' Cyan button	FR-ADAPT-03

Quiz Screen Specification

Element	Value / Behaviour	SRS Ref
Progress bar	'Question X of 10' 11sp Muted; Cyan fill bar; Lime 'Auto-saved' pill top-right	FR-QUIZ-01
Question card	Dark card with Cyan top accent 4dp; question text Poppins Bold 14sp White	FR-QUIZ-01
Answer options	78dp height; selected: Cyan border 1.5dp + 18% cyan tint + Bold White text; left letter block: selected=Cyan 30% fill	FR-QUIZ-01
Next button	Full-width 50dp Cyan; Navy text Poppins Bold 14sp; 10dp radius; arrow icon right	FR-QUIZ-01
Timer	Bottom: 'X/10 answered • Time: X:XX' 11sp Muted centred	FR-QUIZ-01

2.6 Instructor Dashboard and QoE Feedback — Screens 10 and 11



Instructor Dashboard Specification

Element	Value / Behaviour	SRS Ref
Header	Surface 1 96dp; greeting White Bold 14sp; 'Instructor' Indigo pill badge	FR-AUTH-03
Stats cards	3 cards: Students (Cyan), Courses (Lime), Avg Score (Violet) — large bold number + label	FR-ANALYTICS-01
Attention alert	Rose 18% opacity card with Rose left border 4dp; '< 30% completion' students listed with rose badge	FR-ANALYTICS-02
Upload button	Cyan pill button '+ Upload' Navy text — top-right of Courses section	FR-COURSE-01
Course cards	Same dark card style; Indigo thumbnail area with course icon; completion bar Cyan/Rose by performance	FR-ANALYTICS-01

'Needs attention' badge	Rose 22% opacity pill badge on courses below 30% completion	FR-ANALYTICS-02
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QoE Feedback Screen Specification

Element	Value / Behaviour	SRS Ref
Star rating	5 stars; Amber fill for rated; dark border for unrated; tap to rate	FR-QOE-01
Dimension sliders	3 sliders: Cyan track fill; dark background; 0-10 scale; value shown right in Cyan	FR-QOE-01
Comment field	Dark Card bordered textarea 90dp; placeholder Slate; typed text White	FR-QOE-01
Auto-tags	Cyan and Violet pill badges auto-generated: 'Offline mode used', 'N quality switches'	FR-QOE-02
Submit button	Full-width Cyan 50dp Navy text Poppins Bold 14sp	FR-QOE-01
Footer note	'Your feedback trains the Aptiva AI engine' 10sp Muted — reinforces feedback value	FR-QOE-03

Part 3: Flutter Frontend Implementation

3.1 Architecture

Aptiva's Flutter codebase follows **Clean Architecture** with three isolated layers. This separation ensures the business logic is testable independently of the UI, and the data sources can be swapped without touching the screens.

Layer	Responsibility	Location
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Presentation	Screens, widgets, state management (Provider/Riverpod)	lib/screens/, lib/widgets/
Domain	Business logic, use cases, entities — no Flutter dependencies	lib/domain/
Data	REST API clients, SQLite repositories, Firebase calls	lib/data/

3.2 Project Folder Structure

lib/ |—— main.dart ← App entry point; theme injection; route setup |—— app/ | |—— theme.dart ← AptivaCOLORS, AptivaTEXT, ThemeData | |—— routes.dart ← Named routes (GoRouter) | |—— screens/ | |—— auth/ | | |—— splash_screen.dart | | |—— login_screen.dart | | |—— register_screen.dart | |—— student/ | | |—— dashboard_screen.dart | | |—— course_list_screen.dart | | |—— course_detail_screen.dart | | |—— video_player_screen.dart | | |—— audio_only_screen.dart | | |—— offline_screen.dart | | |—— quiz_screen.dart | |—— instructor/ | | |—— instructor_dashboard.dart | | |—— upload_screen.dart | | |—— analytics_screen.dart | | |—— create_quiz_screen.dart | |—— shared/ | |—— qoe_feedback_screen.dart | |—— profile_screen.dart | |—— widgets/ | |—— network_quality_badge.dart ← Most critical widget | |—— course_card.dart | |—— adaptive_player.dart | |—— bottom_nav_bar.dart | |—— audio_waveform_widget.dart ← Animated waveform | |—— shimmer_card.dart | |—— aptiva_button.dart | |—— models/ | |—— user.dart | |—— course.dart | |—— lecture.dart | |—— quiz.dart | |—— quiz_answer.dart | |—— network_status.dart | |—— services/ | |—— api_service.dart ← Dio HTTP client | |—— auth_service.dart ← Firebase Auth wrapper | |—— network_monitor_service.dart ← connectivity_plus + measurements | |—— offline_cache_service.dart ← sqflite CRUD | |—— sync_service.dart ← workmanager background sync

3.3 Flutter Dependencies (pubspec.yaml)

Package	Version	Purpose
google_fonts	^6.1.0	Poppins and Inter typefaces
video_player	^2.8.1	Core video playback engine

chewie	^1.7.4	Video player UI controls (fullscreen, scrubber, controls)
connectivity_plus	^5.0.2	Detect network type — WiFi/4G/3G/2G/none
sqflite	^2.3.0	Local SQLite for offline_cache.db
path_provider	^2.1.1	Device filesystem paths
fl_chart	^0.66.0	Analytics charts: bar, donut, line
provider	^6.1.1	State management — network quality reactive state
dio	^5.4.0	HTTP client with interceptors and upload progress
cached_network_image	^3.3.1	Cache and display course thumbnails
shimmer	^3.0.0	Skeleton loading animation
lottie	^3.0.0	JSON-based animations (waveform, network icons)
flutter_secure_storage	^9.0.0	JWT secure storage — Keychain / Keystore
agora_rtc_engine	^6.3.0	WebRTC live lecture sessions
firebase_core	^2.24.0	Firebase SDK base
firebase_messaging	^14.7.10	FCM push notification receiver
workmanager	^0.5.2	Background sync service — offline queue flush
go_router	^13.0.0	Declarative navigation with deep linking

intl	^0.19.0	Date/time and number formatting
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3.4 theme.dart — Aptiva Design System

The theme.dart file is the **single source of truth** for all visual constants. No colour hex codes, font names, or sizes shall appear anywhere in screen files. All screens import and reference only theme constants.

```
/// Aptiva Design System — theme.dart
class AptivaColors {
  // Primary accents
  static const Color cyan = Color(0xFF00D4FF);
  static const Color indigo = Color(0xFF6366F1);
  static const Color lime = Color(0xFF39D98A);
  static const Color amber = Color(0xFFFFBBF2);
  static const Color rose = Color(0xFFFF43F5);
  static const Color violet = Color(0xFFA855F7);
  // Backgrounds
  static const Color background = Color(0xFF0A0E1A);
  static const Color surface1 = Color(0xFF111827);
  static const Color surface2 = Color(0xFF1A2236);
  static const Color border = Color(0xFF1E2D45);
  // Text
  static const Color textPrimary = Color(0xFFFF8FAFC);
  static const Color textSecondary = Color(0xFF94A3B8);
  static const Color textMuted = Color(0xFF64748B);
}

class AptivaText {
  static const TextStyle display = TextStyle(
    fontFamily: 'Poppins',
    fontSize: 32,
    fontWeight: FontWeight.w700,
    color: AptivaColors.textPrimary,
  );
  static const TextStyle h1 = TextStyle(
    fontFamily: 'Poppins',
    fontSize: 24,
    fontWeight: FontWeight.w700,
    color: AptivaColors.textPrimary,
  );
  static const TextStyle h2 = TextStyle(
    fontFamily: 'Poppins',
    fontSize: 20,
    fontWeight: FontWeight.w600,
    color: AptivaColors.textPrimary,
  );
  static const TextStyle bodyL = TextStyle(
    fontFamily: 'Inter',
    fontSize: 16,
    fontWeight: FontWeight.w400,
    color: AptivaColors.textPrimary,
  );
  static const TextStyle bodyM = TextStyle(
    fontFamily: 'Inter',
    fontSize: 14,
    fontWeight: FontWeight.w400,
    color: AptivaColors.textSecondary,
  );
  static const TextStyle label = TextStyle(
    fontFamily: 'Inter',
    fontSize: 12,
    fontWeight: FontWeight.w500,
    color: AptivaColors.textSecondary,
  );
}

ThemeData aptivaTheme() =>
  ThemeData(
    useMaterial3: true,
    brightness: Brightness.dark,
    scaffoldBackgroundColor: AptivaColors.background,
    colorScheme: ColorScheme.dark(
      primary: AptivaColors.cyan,
      secondary: AptivaColors.indigo,
      error: AptivaColors.rose,
      surface: AptivaColors.surface1,
      background: AptivaColors.background,
    ),
    appBarTheme: AppBarTheme(
      backgroundColor: AptivaColors.surface1,
      foregroundColor: AptivaColors.textPrimary,
      elevation: 0,
    ),
    elevatedButtonTheme: ElevatedButtonThemeData(
      style: ElevatedButton.styleFrom(
        backgroundColor: AptivaColors.cyan,
        foregroundColor: AptivaColors.background,
        minimumSize: const Size(double.infinity, 50),
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(10),
        ),
        textStyle: AptivaText.label.copyWith(
          fontSize: 14,
          fontWeight: FontWeight.w700,
        ),
      ),
    ),
  );
```

3.5 Critical Widget: NetworkQualityBadge

This is the most important widget in Aptiva. It must be implemented first before any screen work begins. It is referenced on the dashboard, the video player, and the audio-only screen.

Band	Badge Label	Badge Color	Icon	Triggers
Excellent (>5Mbps)	Excellent Signal	Lime (#39D98A)	wifi_4_bar	Normal — no action
Good (1– 5Mbps)	Good Signal	Amber (#FBBF24)	wifi	No action
Poor (<1Mbps)	Low Bandwidth	Rose (#F43F5E)	wifi_1_bar	Switch to audio-only (FR-ADAPT-02)
Offline (none)	Offline Mode	Slate (#64748B)	wifi_off	Serve from D6 cache (FR-ADAPT-03)

3.6 Implementation Sequence

Phase	What to Build	Why this order	Est. Time
1 — Foundation	theme.dart, routes.dart, main.dart	No screen works without the design system	1 day
2 — Authentication	Splash, Login, Register screens	Every session starts here	2 days
3 — Student Core	Dashboard, Course List, Course Detail	Core learning flow — most-used path	3 days
4 — Adaptive Engine	NetworkQualityBadge, Video Player, Audio-Only Mode	Heart of the project — adaptive differentiation	4 days
5 — Offline	Offline Screen, Quiz Screen, Background Sync	Validates offline-first architecture	3 days

6 — Instructor	Instructor Dashboard, Upload, Analytics, Create Quiz	Secondary actor — completes the platform	3 days
7 — Shared	QoE Feedback, Profile, Settings	Closes the QoE feedback loop	2 days
8 — Polish	Lottie animations, transitions, error states, accessibility	Production readiness	2 days

Part 4: Aptiva Component Library

4.1 Core UI Components

Component	Dimensions	Variants	Flutter Widget
AptivaButton	Full-width, 50dp height, 10dp radius	Primary (Cyan/Navy), Danger (Rose/White), Ghost (transparent/Cyan border)	ElevatedButton with AptivaTheme style
AptivaTextField	Full-width, 48dp height, 10dp radius	Default (Card bg, dark border), Active (Cyan border 1dp), Error (Rose border)	TextFormField with dark InputDecoration
AptivaCard	Variable width, 12dp radius	Surface2 background, dark border 0.5dp, optional top accent bar	Container with BoxDecoration
CourseCard	Full-width for list / 172dp for grid, 136dp height	With thumbnail, without, shimmer loading	Custom StatelessWidget

NetworkQualityBadge	Variable width, 26dp height, pill radius	4 states: Excellent/Good/Poor/Offline	Custom Widget + AnimatedSwitcher 300ms
AptivaBottomNav	Full-width, 72dp height, Surface1 bg	4 tabs; active = Cyan + top bar 3dp; inactive = Muted	BottomNavigationBar with custom styling
AptivaProgressBar	Full-width, 5-8dp height, 3-4dp radius	Course progress (Cyan), Storage (Cyan), Quiz (Cyan), Audio (Rose)	LinearProgressIndicator
ShimmerCard	Matches card dimensions	Course card shimmer, list item shimmer	shimmer package + Container placeholders
FormatBadge	Variable width, 22dp height, pill	HD (Cyan), SD (Amber), Audio (Rose), Offline (Slate)	Container with border + text
AudioWaveform	Fixed 180dp wide, 40dp height	8 animated bars; colour param; animates to amplitude	Custom AnimatedWidget + Lottie

4.2 Motion and Animation Specification

Animation	Duration	Curve	Trigger	Implementation
Network badge colour change	300ms	Curves.easeInOut	Band changes	AnimatedContainer + color tween
Audio waveform bars	600ms looping	Curves.easeInOut	Audio playing	Custom AnimatedWidget

Screen transition (push)	250ms	Curves.easeInOut	Route navigation	GoRouter with custom PageTransition
Audio-only banner slide-in	400ms	Curves.easeOutCubic	Poor band detected	SlideTransition from top
Card appear on scroll	200ms	Curves.easeOut	ListView item enters view	FadeIn + SlideIn combination
Shimmer sweep	1500ms looping	Linear	Content loading	shimmer package built-in
Play button pulse	800ms looping	Curves.easeInOut	Video paused on splash	ScaleTransition subtle 1.0 → 1.05
Quality switch confirmation	500ms	Curves.elasticOut	Format changes	Brief badge bounce scale effect

4.3 Accessibility Specification

- All interactive elements: minimum touch target 48×48dp — enforced via SizedBox wrappers
- All image/icon elements: semanticsLabel property set for VoiceOver and TalkBack
- Network status communicated via BOTH colour AND text label — never colour alone
- Audio-only mode announced via Semantics.liveRegion for screen reader users
- All form fields: labelText, hintText, and errorText defined for every TextFormField
- Colour contrast verified: Cyan on Navy = 6.8:1 (AAA pass); Rose on Navy = 5.1:1 (AA pass)

- Font minimum: 12sp — enforced in theme.dart; no override permitted
- Focus traversal order tested for keyboard navigation completeness